

The Effect of Calcium Supplementation on the Circadian Rhythm of Bone Resorption

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ABSTRACT

Bone resorption shows a circadian rhythm in human subjects, but the physiological mechanisms underlying this rhythm are unknown. We compared the circadian rhythm of bone collagen degradation in 18 premenopausal women before and after oral calcium supplementation (1000 mg calcium for 14 days). Subjects were randomized to receive calcium at either 0800 h or 2300 h. Continuous 48-h urine collections and 1 day of 4-h urine collections were obtained before and after the 14-day supplementation period. We measured urinary deoxypyridinoline (Dpd) and the cross-linked *N*-telopeptide of type I collagen (NTx) as biochemical markers of bone resorption. There was a significant effect of time of day on excretion of Dpd and NTx (analysis of variance, $P < 0.001$) with peak excretion between 0300–0700 h and a nadir between 1500–1900 h. The mean amplitude (peak to trough) was similar for Dpd and NTx (70.3% and 63.3%, respectively). Evening calcium

supplementation resulted in marked suppression of the nocturnal increase in Dpd and NTx and reversed the usual nocturnal increase in the level of parathyroid hormone. In contrast, morning calcium supplementation had no significant effect on the circadian rhythm of Dpd or NTx. Evening calcium supplementation suppressed overall daily excretion of Dpd by 20.1% ($P = 0.03$) and NTx by 18.1% ($P = 0.03$). Morning calcium supplementation had no significant effect on overall daily excretion of either Dpd or NTx. We conclude that evening calcium supplementation suppresses the circadian rhythm of bone resorption. The daily rhythm of PTH secretion or calcium intake is likely to be an important determinant of this rhythm. Experimental protocols designed to investigate the effect of calcium supplementation on bone mineral density should take the timing of supplementation into account. (*J Clin Endocrinol Metab* 79: 730–735, 1994)

A MARKED circadian rhythm of bone turnover has been demonstrated in several mammalian species using a variety of techniques including ^{45}Ca kinetic studies (1, 2), excretion of [^3H]tetracycline from prelabeled animals (3), measurement of [^3H]proline and [^3H]thymidine metabolism in rat bone (4), and excretion of hydroxyproline in urine (5). Several new and potentially specific assays for the estimation of bone resorption have been developed during recent years, especially those relating to the degradation of type I collagen. The collagen cross-link deoxypyridinoline (Dpd) is found almost exclusively in type I collagen of bone and is not absorbed from the diet (6). Studies in healthy human subjects have demonstrated a large-amplitude circadian variation in excretion of Dpd under conditions of nutritional and sleep-time synchronization (7, 8, 9, 10). Excretion of Dpd is highest at around 0700 h and lowest around 1700 h. Although markers of bone formation also show a circadian rhythm, the relative amplitude of this rhythm is much smaller (8), and it is therefore likely that there is net bone resorption at night. Temporal changes in other novel markers of bone resorption, such as the *N*-telopeptide cross-linking domain of type I collagen (NTx) in urine (11), have not been studied.

The physiological basis for the circadian rhythm of bone resorption is uncertain. In general, observed circadian rhythms can be described in terms of two components: 1) an endogenous component controlled by a body clock; and 2) an exogenous component that results from cyclical changes in the external environment. These components are normally in phase, and exogenous influences have the effect of amplifying the endogenous rhythm. Although it has been proposed that endogenous factors may be important in controlling some aspects of the circadian rhythm of calcium homeostasis (12, 13, 14, 15), there are also several possible exogenous modulators of bone resorption, including the daily rhythm of posture, motor activity, and dietary calcium intake. Although bed rest results in increased bone resorption within a few days (16), the circadian variation in Dpd excretion is not altered by recumbent posture in human studies (17). There is some evidence that in laboratory animals, the circadian rhythm of [^3H]tetracycline excretion and ^{45}Ca kinetics are related to the daily rhythm of calcium intake, or the intake of other nutrients (2, 3). The rhythm of bone resorption could also depend on a number of endocrine circadian rhythms. Several hormones such as cortisol, PTH, and calcitonin modulate bone turnover, and these hormones show circadian rhythmicity. Under normal conditions, levels of PTH show a circadian rhythm, with minimal concentrations around 1000 h and maximum levels at about 0300 h (14, 18–21).

Calcium supplementation may prevent bone loss in postmenopausal women (22) and has been shown to reduce bone

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resorption in postmenopausal women, as reflected by a reduction in fasting hydroxyproline excretion (23, 24). The effect of calcium supplements on bone turnover in premenopausal women using bone-specific markers of collagen degradation is not known.

The aims of this study were: 1) to compare the circadian rhythm of bone collagen degradation as determined by urinary excretion of Dpd and NTx in healthy premenopausal women; 2) to assess whether the nocturnal increase in the level of PTH is related to the circadian rhythm of bone resorption by studying the effect of timed oral calcium supplementation; and 3) to assess the effect of timed calcium supplementation on overall daily bone resorption in premenopausal women.

Subjects and Methods

Subjects

We studied 18 healthy premenopausal white women 20–47 yr of age (mean age, 36 yr). Two subjects had used oral contraceptives within the previous 3 months, and all had regular menstrual cycles. None had any known systemic disease or was taking any medication known to influence calcium homeostasis. Subjects were excluded if they had recently traveled across several time zones or had suffered major fractures within the previous 2 yr. All subjects had regular sleep-wake cycles roughly consistent with the environmental light-dark cycle. During the week before each study, subjects were asked to adhere to a schedule of sleep and wake times as close as possible to those in the circadian study. Subjects were asked to avoid excessive alcohol intake or intensive exercise for the entire study period. No other attempts were made to modify the subjects' usual pattern of dietary intake. One week before the study, each subject completed a 4-day weighed dietary record administered by a trained dietitian. The 4-day average calcium intake was determined using a computer package (Microdiet, University of Salford, Salford UK). The studies were approved by the Northern General Hospital ethics committee, and signed informed consent was obtained from all subjects.

Study protocol

Circadian studies were performed before and after 2 weeks of oral calcium supplementation. After the first circadian study, each subject received 1000 mg elemental calcium/day as calcium citrate (Cacit, Proctor and Gamble Ltd., Staines, Middlesex, UK) for 14 days, including the second circadian study day. Subjects were randomly assigned to receive the supplement at either 0800 h ($n = 9$) or 2300 h ($n = 9$).

Subjects were admitted to the Bone and Joint Research Unit, University of Sheffield, at 1800 h on each study day. An intravenous cannula was inserted for collection of blood samples. Blood was drawn every 2 h from 1900 h until 1900 h the following day. Samples were allowed to clot for 30 min, centrifuged at 3000 rpm for 6 min, and aliquots of serum were stored at -80°C until analyzed. The total volume of blood sampled was approximately 200 mL/24 h. Urine was collected during the 24-hour period of each study in six separate 4-h collections starting at 1900 h. Exact times of bladder emptying and the volume of each collection were recorded. Portions of the urine samples without preservative were stored at -80°C until analyzed. In addition to the urine samples provided during the circadian studies, each subject collected two consecutive 24-h urine samples immediately before each circadian study.

Subjects were kept under close staff supervision at all times, and posture and meal timing were strictly controlled. Subjects were recumbent (head and limbs at the same level) with the lights off from 2300–0700 h. During the dark period, a weak (3-watt) spotlight was used to facilitate the collection of blood samples. Subjects were awake and ambulant (sitting or walking) from 0700–1900 h and were not permitted to lie down or to sleep. Meals were provided at exactly 1930 h, 0730 h,

and 1330 h and were consumed within 30 min. Tea and a biscuit were provided at 2100 h, 1030 h, and 1630 h. On the first study day, meals were self-selected from a range of prepared foods, and subjects were asked to select foods that they would consume during a typical day. The self-selected foods were weighed, and subjects were provided with an identical diet on the second study day. No attempt was made to study subjects at the same phase of the menstrual cycle. Studies in subjects randomized to receive morning and evening calcium supplementation were performed concurrently during a period of 1 yr.

Biochemical methods

All samples from an individual were analyzed in duplicate in the same assay to minimize interassay variation. We measured total Dpd and pyridinoline in urine by high-performance liquid chromatography (HPLC) after acid hydrolysis using the method of Colwell *et al.* (6). Complete baseline separation of pyridinoline and Dpd was achieved for all samples. Quantification was achieved using a secondary external standard purified from urine and calibrated against a standard obtained from Dr. S. Robins (Rowett Institute, Aberdeen, UK). The intraassay coefficient of variation (CV) was 5%, and the interassay CV was 8%. Sample preparation as well as HPLC were performed in duplicate. If the difference between assay duplicates differed by more than 12%, all the samples collected from that subject were reanalyzed.

NTx was measured using a recently developed enzyme-linked immunosorbent assay (Osteomark, Ostex International, Seattle, WA). This assay may offer several practical advantages over the measurement of Dpd by HPLC and has been described in detail elsewhere (11). The results were expressed as equivalent moles of bone type I collagen (bone collagen equivalents) normalized to creatinine (Cr) (11). The intraassay CV was 8%, and the interassay CV was 10%.

Intact PTH was determined using an immunoradiometric assay (Allegro PTH, Nichols Institute, San Juan Capistrano, CA). The intraassay CV was 3%, and the interassay CV was 6%. Urinary Cr was measured using a kinetic Jaffé reaction.

Statistical methods

All data are expressed as mean $\pm$ SEM. Observations at concurrent time points were averaged, and the presence of a time-of-day effect was determined by two-factor analysis of variance (ANOVA). Dpd and NTx showed a skewed distribution, and data were logarithmically transformed. To avoid the influence of the substantial between-subject variability in the mean level of Dpd and NTx, data were also expressed as a percentage of the individual 24-h mean value. The effect of morning or evening calcium supplementation was analyzed by applying two-tailed paired *t* tests at all time points. The effect of morning or evening calcium supplementation on overall daily bone resorption was determined by comparing the 48-h mean excretion of each marker before and after calcium supplementation using a paired *t* test. The effect of timing of calcium supplementation on the change in overall excretion of Dpd and NTx was determined using an unpaired *t* test. All statistical tests were two sided, and $P < 0.05$ was regarded as significant. The effect of time of day and calcium supplementation on excretion of NTx and Dpd were similar whether results were expressed as a rate of excretion or as a ratio to Cr. Therefore, all urinary data are expressed as ratios to Cr excretion.

Results

There was no significant difference between the groups randomized to receive calcium supplementation in the morning or evening, with respect to age, weight, height, or habitual calcium intake (Table 1). Forty-eight-hour excretion of Dpd and NTx determined before calcium supplementation was higher in the group randomized to receive evening calcium supplementation, although this was not statistically significant (Table 1).

Figure 1 shows the circadian variation in Dpd and NTx in

TABLE 1. Baseline characteristics of subjects randomized to receive calcium supplementation at 0800 h (n = 9) and 2300 h (n = 9)

Variable	0800-h group (mean $\pm$ SEM)	2300 h group (mean $\pm$ SEM)	P
Age	33.6 $\pm$ 4.2	32.2 $\pm$ 3.3	0.77
Weight (kg)	63.0 $\pm$ 5.3	67.6 $\pm$ 3.8	0.19
Height (cm)	157 $\pm$ 1.3	162 $\pm$ 2.9	0.33
Calcium intake (mg/day)	847 $\pm$ 98	1129 $\pm$ 126	0.09
NTx/Cr (μ mol bone collagen equivalents/mol) ^a	45.0 $\pm$ 10.8	56.8 $\pm$ 7.2	0.45
Dpd/Cr (μ mol/mol) ^a	6.2 $\pm$ 1.2	9.6 $\pm$ 2.1	0.20

^a Determined from 48-h urine collection before the first circadian study.

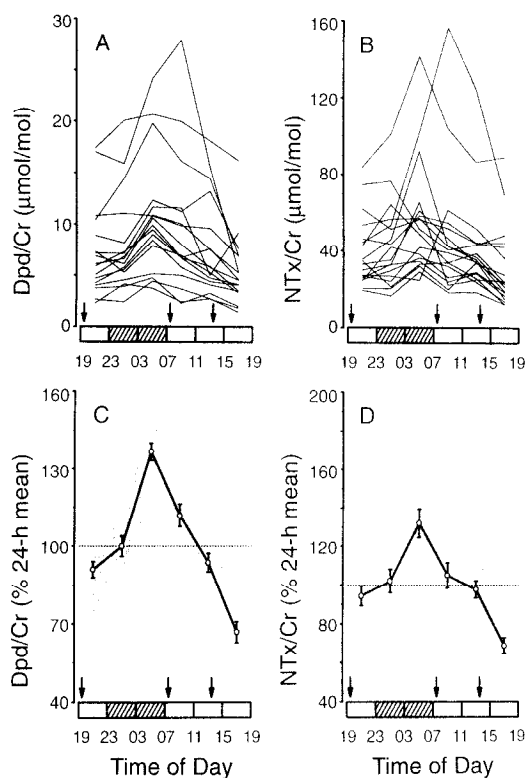


FIG. 1. Circadian variation in excretion of Dpd and NTx in all subjects (n = 18) before calcium supplementation. A and B, Measured Dpd and NTx excretion for individual subjects. C and D, Percentages of the 24-h mean excretion in each subject. Each thin line represents one individual. The solid lines represent the mean $\pm$ SEM of the normalized values. Data are plotted at the midpoint of each 4-h collection period. Subjects were recumbent with the lights off during the period indicated by the shaded bar on the horizontal axis. Meals were taken at the times indicated by arrows. There was a significant effect of time of day for both variables (ANOVA, $P < 0.001$).

all subjects before calcium supplementation. There was a significant effect of time of day on the excretion of Dpd and NTx (ANOVA, $P < 0.001$) with peak excretion between 03.00 and 07.00 h and a nadir between 15.00 and 19.00 h (Fig. 1). The form of the circadian variation was essentially the same for Dpd and NTx. The mean amplitude (peak to trough) was 70.3% for Dpd and 63.3% for NTx. Inspection of curves for individual subjects reveals a large heterogeneity in the form

of the rhythm. The somewhat greater heterogeneity in the form of the NTx rhythm compared with that of Dpd may be accounted for by the stringent protocol that was used to determine the adequacy of within-run analytical precision in the Dpd assay (see *Subjects and Methods*).

Evening calcium supplementation suppressed the nocturnal increase in excretion of Dpd and NTx (Fig. 2, A and C). There was no significant effect of time of day on the excretion rate of Dpd and NTx after evening calcium supplementation. In contrast, there was no significant change in the circadian variation of either Dpd or NTx in subjects given morning calcium supplementation (Fig. 2, B and D; ANOVA, $P < 0.001$).

Before calcium supplementation, we observed a nocturnal increase in the levels of PTH in both groups of subjects (Fig. 3). Evening calcium supplementation reversed the nocturnal rise in PTH but did not affect PTH levels during the day. In contrast, there was no significant change in nocturnal PTH levels in the group given morning calcium supplementation, although PTH levels were suppressed during the day.

The effect of timing of calcium supplementation on overall daily excretion of Dpd and NTx (determined from the 48-h urine collection before and after calcium supplementation) is shown in Figs. 4 and 5. Evening calcium supplementation reduced overall daily excretion of Dpd by 20.1% ($P = 0.03$,

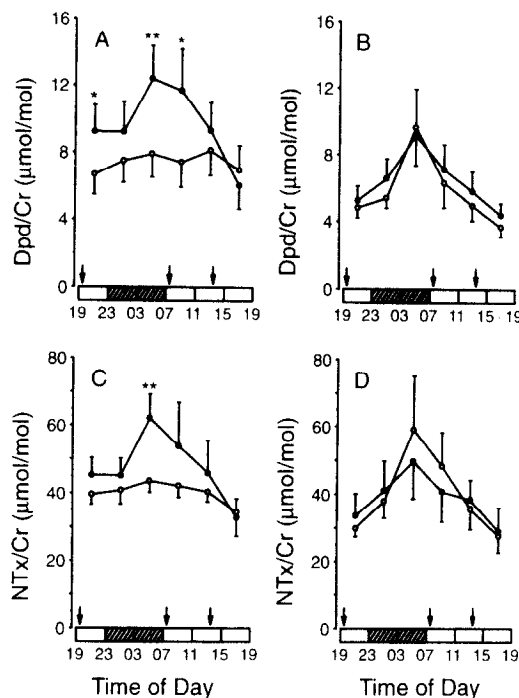


FIG. 2. Effect of timed calcium supplementation on the circadian rhythm of Dpd (upper panels) and NTx (lower panels). A and C, Circadian rhythm on the control day (●) and after 1000 mg calcium given at 2300 h (○). B and D, Circadian rhythm on the control day (●) and after 1000 mg calcium given at 0800 h (○). Values are the mean $\pm$ SEM of nine subjects for each cycle. The shaded bar on the horizontal axis represents the period of darkness. Data are plotted at the times indicated by arrows. *, $P < 0.05$, paired t test vs. control cycle; **, $P < 0.01$, paired t test vs. control cycle.

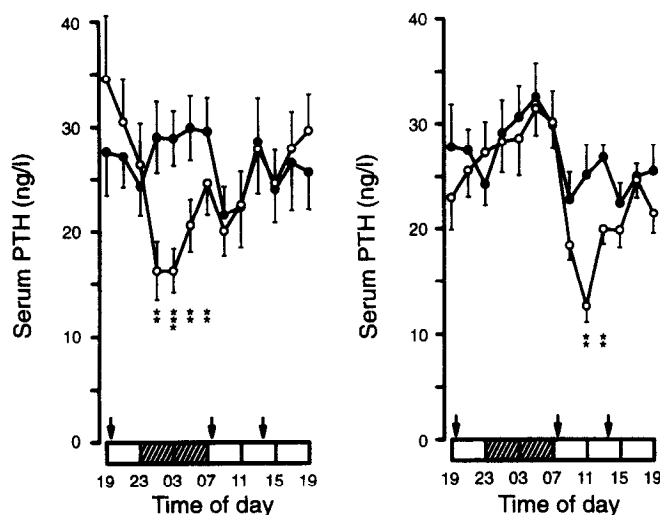


FIG. 3. Effect of timed calcium supplementation on the circadian rhythm of PTH. *Left panel*, Circadian rhythm on the control day (●), and after 1000 mg calcium given at 2300 h (○). *Right panel*, Circadian rhythm on the control day (●), and after 1000 mg calcium given at 0800 h (○). Values are the mean $\pm$ SEM of nine subjects for each cycle. The shaded bar on the horizontal axis represents the period of darkness. Meals were taken at the times indicated by arrows. There was a significant effect of time of day on the level of PTH in both groups of subjects both before and after calcium supplementation (ANOVA, $P < 0.001$). **, $P < 0.01$, paired t test vs. control cycle; ***, $P < 0.001$, paired t test versus control cycle.

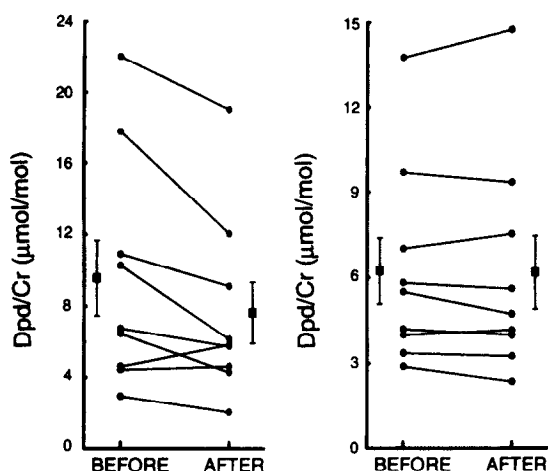


FIG. 4. Individual changes in overall daily excretion of Dpd after calcium supplementation for 14 days. Mean 48-h excretion of Dpd before and after calcium administration at 2300 h (*left panel*) or 0800 h (*right panel*). Each horizontal line represents one individual. Dpd/Cr was reduced by 20.1% by evening calcium supplementation (paired t test, $P = 0.03$), but morning calcium supplementation had no significant effect.

paired t test; 95% confidence interval for mean change in Dpd/Cr, -3.53 to -0.31 $\mu\text{mol/mol}$) and NTx by 18.1% ($P = 0.03$, paired t test; 95% confidence interval for mean change in NTx/Cr, -19.3 to -1.2 $\mu\text{mol/mol}$). Evening calcium supplementation resulted in a decline in 48-h Dpd/Cr in seven of nine subjects and a decline in NTx/Cr in eight of nine subjects.

Morning administration of calcium did not significantly

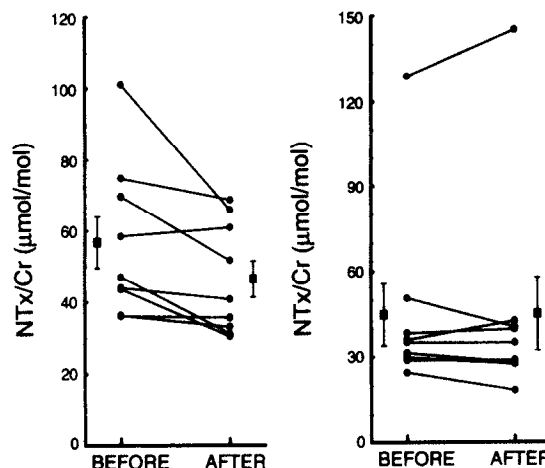


FIG. 5. Individual changes in overall daily excretion of NTx after calcium supplementation for 14 days. Mean 48-h excretion of NTx before and after calcium administration at 2300 h (*left panel*) or 0800 h (*right panel*). Each horizontal line represents one individual. NTx/Cr was reduced by 18.1% by evening calcium supplementation (paired t test, $P = 0.03$), but morning calcium supplementation had no significant effect.

alter overall excretion of either Dpd or NTx (95% confidence interval for mean change in Dpd/Cr, -0.46 to -0.38 $\mu\text{mol/mol}$; 95% confidence interval for mean change in NTx/Cr, -5.5 to 6.2 $\mu\text{mol/mol}$). The effect of morning calcium supplementation on 48-h excretion of Dpd and NTx remained nonsignificant when the single subject with the highest basal excretion of Dpd and NTx was excluded from the analysis. However, given the relatively small number of subjects in each group, it was not possible to show that the decrease in overall bone resorption was significantly greater for evening than for morning supplementation. There was no significant relationship between habitual dietary calcium intake, and the change in excretion of either Dpd or NTx after calcium supplementation.

Discussion

In this study, we have confirmed the previous finding of a marked diurnal rhythm of Dpd excretion in healthy women under conditions of nutritional and sleep-time synchronization, with increased levels at night and a nadir in the later afternoon (7–10). Furthermore, we have demonstrated that there is a similar rhythm in excretion of NTx. These findings in premenopausal women support earlier studies in postmenopausal women showing that hydroxyproline excretion in early-morning urine samples is reduced by evening calcium supplementation (23, 24).

We have shown that oral calcium supplementation given at 2300 h reverses the nighttime increase in PTH and attenuates the increase in bone resorption. Overall daily excretion of Dpd and NTx decreased significantly in subjects given evening calcium supplementation but did not decrease in subjects given the supplement in the morning. It was, however, not possible to show that the decrease in overall excretion of Dpd or NTx was significantly greater in subjects given evening rather than morning supplementation, and a larger

study will be required to demonstrate if this is indeed the case.

These findings support earlier animal studies showing that the circadian rhythm of serum total Ca (25), ^{45}Ca kinetics (2), and $[^3\text{H}]$ tetracycline excretion (3) are modulated by dietary intake of calcium or other nutrients. However, these techniques reflect bone mineral kinetics and may not necessarily reflect bone collagen degradation (26). Minor changes in extracellular calcium concentration are buffered by exchangeable pools in bone (27), and the serum activity of ^{45}Ca at any given time is determined by the combined effects of calcium buffering and true osteoclastic bone resorption. It is clear from human studies that most unidirectional transfer of ^{45}Ca from bone represents efflux from surface exchangeable pools rather than bone resorption (28). The $[^3\text{H}]$ tetracycline model may be more useful than the ^{45}Ca technique, because a proportion of ^{45}Ca that is released from bone is recycled back into bone, interfering with the assessment (3). In human studies, about half the total calcium in serum is protein bound, and the circadian variation in total calcium is simply due to postural variations in the serum protein concentration (29).

It is possible that the circadian rhythm in the level of PTH mediates the rhythm of bone collagen degradation. Infusion of PTH(1–34) stimulates bone resorption and increases urinary excretion of hydroxyproline within 4 h (30). Morphological studies in rats have shown that PTH administration increases osteoclast ruffled border within 30 min, that calcium replenishment of fasted animals reduces osteoclast ruffled border within as little as 3 h, and that activation of recently inhibited osteoclasts occurs within 12 h of fasting (31).

The subjects in this study were treated with pharmacological doses of calcium, and therefore these data do not allow us to ascertain whether physiological variations in the timing of calcium intake have any effect on the circadian rhythm of PTH or bone resorption. Although evening calcium supplementation reverses the nocturnal increase in PTH, the circadian rhythm of Dpd and NTx excretion is attenuated rather than reversed. Furthermore, the suppression of PTH after morning calcium supplementation was not associated with a reduction in morning Dpd or NTx excretion. Therefore, PTH is probably not the only factor determining the circadian rhythm of bone resorption. It is possible that the timing of the rhythm could be related to other factors such as cortisol (32). The effect of calcium supplementation on the circadian rhythm of bone resorption could also be due to a calcium-induced increase in calcitonin (CT) secretion. However, it is not clear whether physiological levels of CT have any important role in regulating calcium homeostasis in humans (33), and calcium supplementation seems to have no effect on CT secretion in healthy premenopausal women (34).

The mechanisms underlying the circadian rhythm in the level of PTH is uncertain. In several studies, an inverse correlation has been demonstrated between circadian changes in serum ionized calcium (iCa) and PTH (18, 19, 21). It is therefore likely that a small decrease in iCa during the nocturnal calcium fast is responsible, at least in part, for

the nighttime increase in PTH. However, there is some confusion in the literature as to the normal circadian pattern of iCa in healthy subjects. These differences may relate to protein interference at the reference-electrode liquid junction of some ion-selective electrodes (35). Free-running circadian oscillations have also been implicated as at least one of the factors controlling the diurnal pattern of PTH secretion. A recently published preliminary study (12) examined the circadian rhythm of PTH in subjects kept on a constant routine for 40 h and showed that the synchronized nocturnal increase in PTH was preserved. This apparent endogenous component would reinforce the exogenous mechanisms when calcium intake is taken at the customary hours. If there is an endogenous element in the control of PTH secretion, this can clearly be overridden by pharmacological manipulation of calcium intake. We have proposed that the nocturnal increase in bone resorption may be required for maintenance of serum calcium during the nocturnal calcium fast (8), and an endogenous rhythm of PTH secretion could therefore be considered to be a mechanism of anticipatory homeostasis (36).

No correlation between age and the relative amplitude of the rhythm of Dpd or NTx was found in this study, although the age range was too narrow and the size of the group too small to exclude such a possible correlation. The individual responses were highly variable, however, and younger subjects (20–24 yr) tended to have higher overall daily resorption rates (48-h means) than older subjects. In a previous study we found no apparent effect of age on the day to night ratio of Dpd excretion in women between 10–75 yr of age (8). Calcium homeostasis in pubertal subjects, postmenopausal women, or subjects with low habitual calcium intake may be more sensitive to dietary calcium intake. A potentially confounding factor in this study was that habitual calcium intake was higher in the group randomized to receive evening calcium supplementation, although this difference was not significant. It is possible that higher habitual calcium intake could result in a greater response to calcium supplementation on the basis of a threshold effect. However, it seems more likely that higher habitual calcium intake would reduce the response to supplementation due to a lower relative increase in calcium intake and net calcium absorption. It was also not possible to study all subjects at the same phase of the menstrual cycle, although levels of bone-specific markers of collagen degradation do not seem to show any significant variation during the menstrual cycle (37).

In conclusion, we suggest that evening calcium supplementation: 1) abolishes the nighttime increase in levels of PTH; 2) attenuates the circadian rhythm of bone resorption; and 3) reduces overall daily bone resorption in healthy premenopausal women. These results suggest that the daily rhythm of bone resorption is determined, at least in part, by the daily rhythm of calcium intake or PTH secretion. Further studies are required to investigate the effect of timing of long-term calcium supplementation on bone turnover and bone mineral density in postmenopausal women. We also suggest that experimental protocols designed to investigate the effect of calcium supplementation on bone mineral den-

sity should take the timing of supplementation into account to avoid ambiguous and possibly misleading conclusions.

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